



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143

PROBABILITY & STATISTICAL DATA ANALYSIS

ASSIGNMENT 1

LECTURER:

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SEMESTER II (2023/2024)

1. (a) Population: students of socioeconomic faculty in the university

Sample: ~~set~~ subset of selected student of socioeconomic faculty in the university

(b) (i) Discrete: number of subjects taken by the student
continuous: GPA of the student

(ii) Categorical: Gender of the student

Numerical: number of hours taken to study every day

(iii) Level of measurement

nominal: Gender - have classification

ordinal: Grade of each subject - have classification and ranked

interval: GPA of the ~~set~~ student - no true zero point

Ratio: time taken to study in a day - have true zero

Question 1.

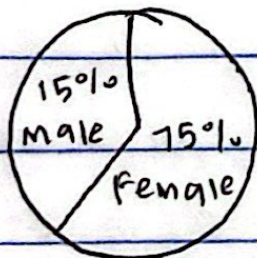
(c) Primary Source Data 3, 3, 1, 1, 2, 2, 2, 1, 1, 1

⇒ study (survey) distributed to the faculty students.

(d) Secondary Data Source 1, 1, 2, 2, 2, 2, 2, 1, 1, 1

Existing
⇒ Records of the survey from university's Faculty.

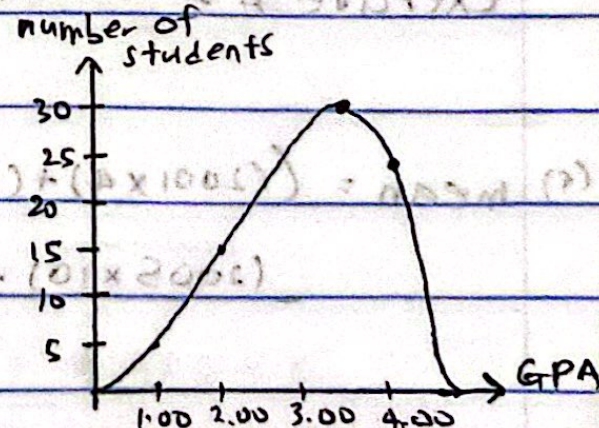
(e) (i) Student's gender



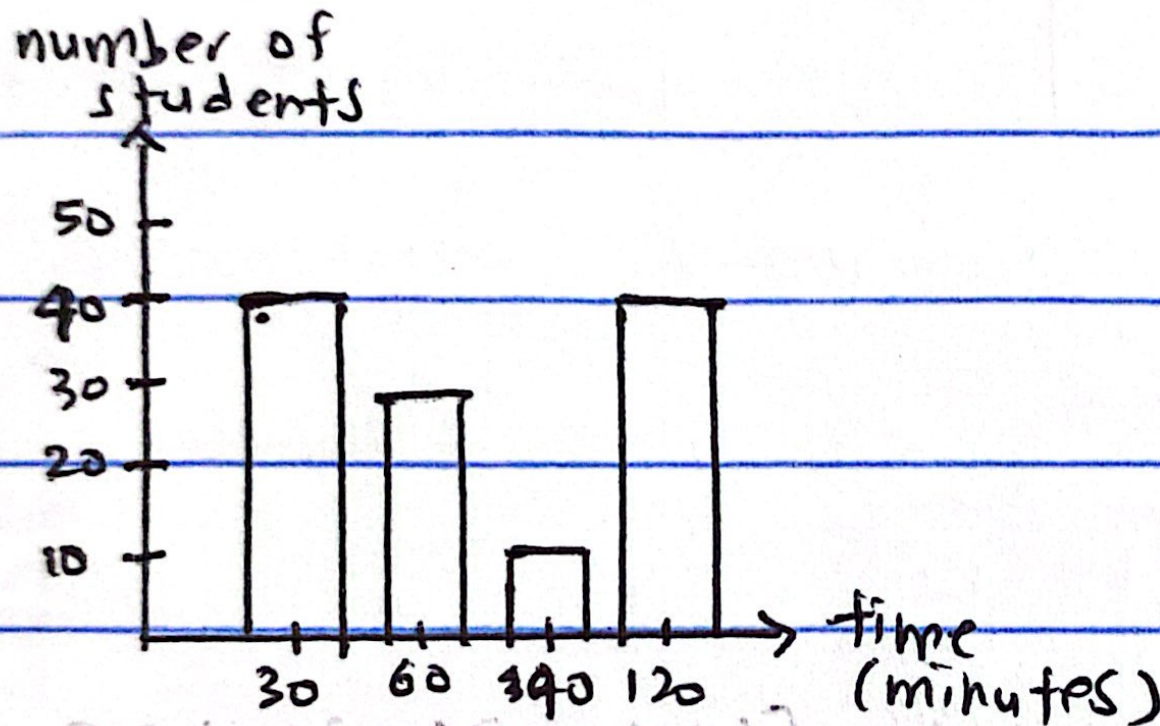
(ii) Grade of each subject

subject	Highest Mark
PSDA	A+
System Analysis & design	A
Human Computer Interaction	A+

(iii) student's GPA



(iv) Time taken to study in a day.



Question 2.

Number of students	Frequency	relative Frequency
0	3	$3 \div 30 = 0.10$
1	2	$2 \div 30 = 0.07$
2	7	$7 \div 30 = 0.23$
3	7	$7 \div 30 = 0.23$
4	5	$5 \div 30 = 0.17$
5	4	$4 \div 30 = 0.13$
6	2	$2 \div 30 = 0.07$
Total	30	1.00

Q3

label weight frequency

450

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451

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452

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453

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454

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455

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456

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468

469

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470

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471

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472

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473

|

Company goals : $450 \leq \text{packages} \leq 458$
weight

■ find Range

$$R = X_h - X_l$$

$$= 473 - 450$$

$$= 23$$

■ trial and error

$$h = R = 23$$

$$\frac{1}{1} \quad \frac{2}{3}$$

$$= 7.67$$

$$\approx 8$$

■ find cell interval

$$i = \frac{R}{1 + 3.322 \log n}$$

$$= \frac{23}{1 + 3.322 \log 100}$$

$$= 3.008895866$$

$$\approx 3.01$$

$$\approx 3$$

■ find midpoint

$$MP_L = X_L + \frac{i}{2}$$

$$= 450 + \frac{3}{2}$$

$$= 451.5$$

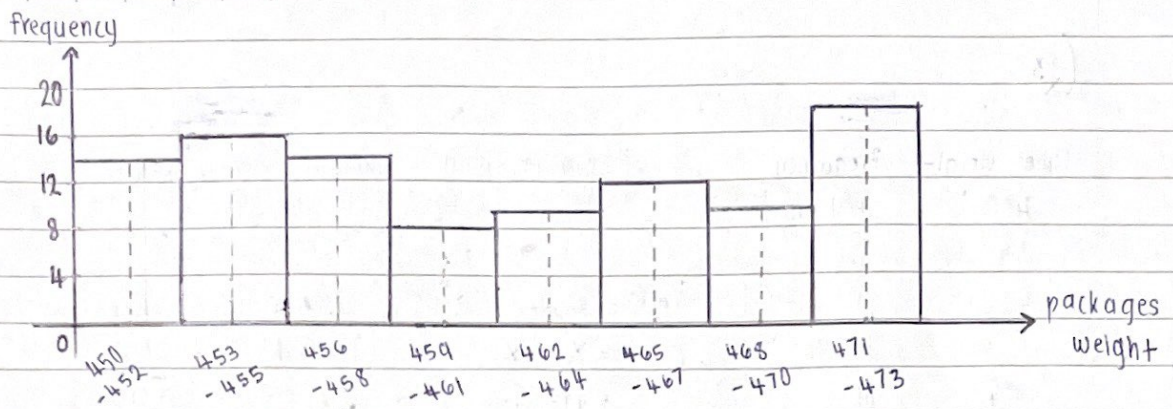
$$\approx 451$$

weight	cell midpoint	cell boundaries	frequency
450 - 452	451	449.5 - 452.5	14
453 - 455	454	452.5 - 455.5	16
456 - 458	457	455.5 - 458.5	14
459 - 461	460	458.5 - 461.5	8
462 - 464	461	461.5 - 464.5	9
465 - 467	466	464.5 - 467.5	12
468 - 470	469	467.5 - 470.5	9
471 - 473	472	470.5 - 473.5	18

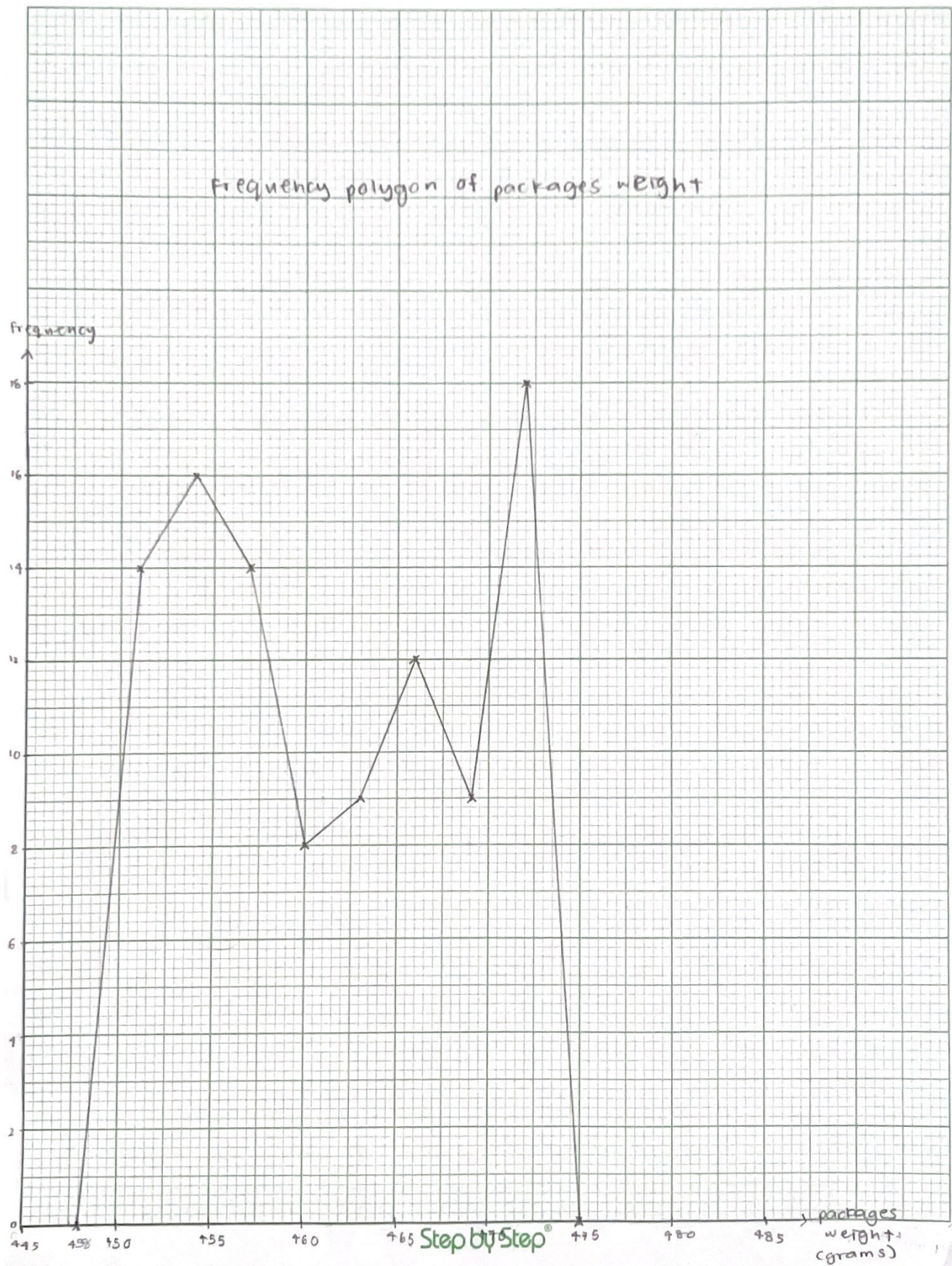
lower bound upper bound

NO. _____

DATE _____



b) frequency polygon



No:

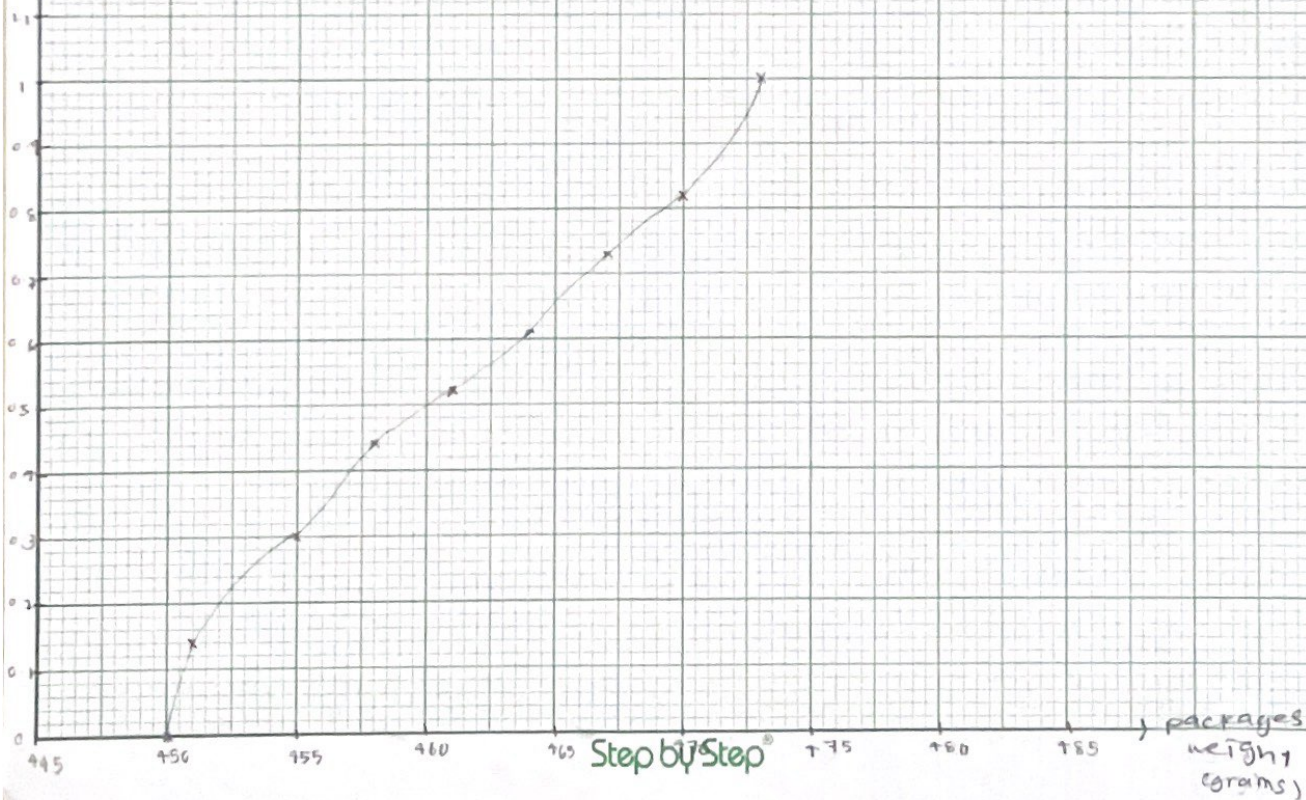
Date:

c)	weight (gram)	cell midpoint	cell boundaries	frequency	relative frequency	cumulative frequency	cumulative relative frequency
	450 - 452	451	449.5 - 452.5	14	0.14	14	0.14
	453 - 455	454	452.5 - 455.5	16	0.16	30	0.30
	456 - 458	457	455.5 - 458.5	19	0.19	49	0.49
	459 - 461	460	458.5 - 461.5	8	0.08	57	0.57
	462 - 464	463	461.5 - 464.5	7	0.07	64	0.64
	465 - 467	466	464.5 - 467.5	12	0.12	76	0.76
	468 - 470	469	467.5 - 470.5	9	0.09	85	0.85
	471 - 473	472	470.5 - 473.5	18	0.18	103	1.00

c) ogne

Relative ogne of packages weight

Cumulative
relative
frequency





Date: _____

4. 78, 702, 765, 811, 832, 855, 896, ~~915, 919, 920~~, 902, 905
918, 919, 920, ^{Q1}923, ~~936, 938~~, 929, 936, 938, 948, 950,
956, ~~958, 959~~, 970, 972, 978, 1009, ^{Q2}1022, 1035, 1037, 1045,
1067, 1085, 1092, 1102, 1122, 1126, 1151, ^{Q3}1156, 1157, 1157,
1162, 1170, 1195, 1195, 1196, 1217, 1237, 1311, 1333, 1340

Min = 78

Q1 = 923

Q2 = 1022

Q3 = 1156

Max = 1340

$$IQR = Q3 - Q1 = 1156 - 923 = 233$$

$$\text{Lower limit} = 923 - 1.5 \times 233 = 573.5$$

$$\text{Upper limit} = 1156 + 1.5 \times 233 = 1505.5$$

78 is lower than the lower boundary, thus make it the outlier. The new min is 702

There is no value surpass the upper boundary.

outlier : 78

